G H Raisoni International Skill Tech University, Pune

Established by Maharashtra Act No. IX of 2024 and under section 2(f) of the UGC Act 1956

End Semester Examinations - Winter Term 2025

School of Engineering and Technology

S.Y. B.Tech., Semester - III (All Branches)

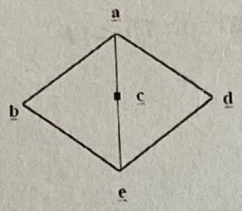
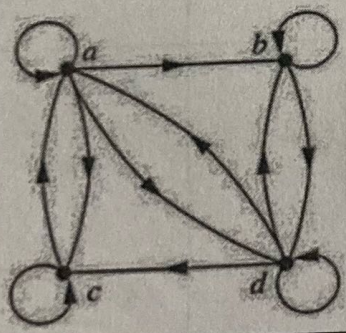
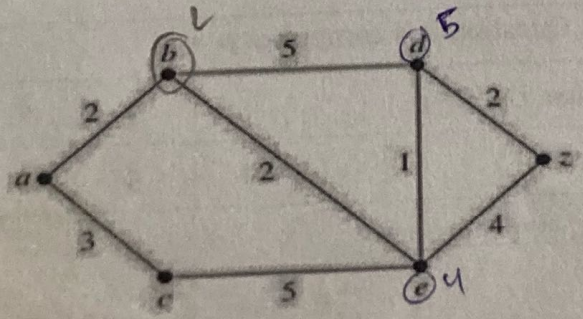
Course Name: Discrete Mathematics and Graph Theory **Course Code: 25BTBSL301**

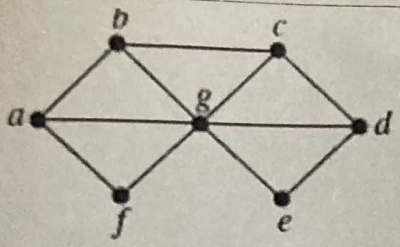
Max Marks: 60 **Duration: 150 minutes**

Instructions:

1. **Write the solution in detail.**
2. Clearly show all steps of your work, including necessary derivations and calculations.
3. Read each question carefully before attempting.
4. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q. No. 2C, etc.
5. Each question is of 12 marks, from every question solve any **TWO from A, B & C**. Question **D** is **compulsory**.

Q. No.	Sub Q.No.	Question	Marks	CO	BL
1		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Let $A = \{1,2,3,4\}$ *and let* $R = \{(1,1),(1,2)(2,1),(2,2),(3,4),(4,3),(3,3),(4,4)\}$ Prove that R is an equivalence relation on A	5	1	2,3
	B	(1,2,3,4,5) Let $A = \{1,2,3,4\}$ and the relation matrix is $\begin{bmatrix}1&1&1&0&0\\1&1&1&0&0\\1&1&1&0&0\\0&0&0&1&1\\0&0&0&1&1\end{bmatrix}$ Find Transitive Closure of R.	5	1	2
	C	Draw a Hasse diagram for divisibility on the set {1,2,3,4,5,6,7,8}	5	2	2,3
	D	Define Power set, for $A = \{1,2,3,4\}$ *Find* $P(A)$.	2	2	1-define
2		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	Let G = Z, show that G is group under addition. i.e.(Z, +)	5	2	3
	B	Compute the product of (1,2,5,7)o(1,3,2) o(2,4,5,6)	5	2	2
	C	Show that $S(n) = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ using mathematical induction	5	2	3
	D	Find generator of Z_7 in under addition modulo 7.	2	2	2

3		Solve any **TWO from Question A, B & C. Question D is compulsory.**			
	A	*Let* $A\{1,2,3,4\}$ *Find* $P(A)$. *i. e. Find Collection of all subsets of s* And draw a Lattice of $(P(A), \subseteq)$.	5	2	2,3
	B	Show that set of all divisors of 48 for divisibility relation forms a lattice. Also find maximal element and minimal element of lattice	5	2	2,3
	C	The following is the Hasse diagram of the POSET $\{\{a,b,c,d,e\}, <$ $\}$. Is it a Lattice? Justify? (Find Composition table) 	5	2	2
	D	Define Integral domain with example.	2	2	1
4		Solve any **TWO from Question A, B & C. Question D is compulsory.**			
	A	Let L be the set of all factors of 12 and '|' be divisibility relation on L. Show that $(L, \|)$ is a lattice.And write the composition table of 'meet' and 'join'	5	3	2
	B	Find the in-degree and out-degree of each vertex in the graph G. Also find Adjacency matrix of G. 	5	3	2
	C	Find the length of a shortest path between a and z in the given weighted graph by using Dijkstra's algorithm. 	5	3	3

	D	 What will be chromatic number of given graph?	2	3	1
5		Solve any **TWO from Question A, B & C**. Question **D is compulsory.**			
	A	**In a class of 100 students,50 students** know English,**45** know French, and **38** know German language.**27 students** know English and French, **24 students** know French and German,**18** know English and German, and **23 students** know none of the languages. Find out: (i) How many students know **all the 3 languages**? (ii) How many students know **exactly 2 languages**? (iii) How many students know **only one language**?	5	4	2,3
	B	How many **4-digit numbers** can be formed using the digits 0–9 if: (i) Repetition is allowed (ii) Repetition is not allowed (iii) The number must be an **even number**?	5	4	3
	C	Prove that in any group of **367 people**, at least **two people** must have the **same birthday**. Use the generalized pigeonhole principle and explain clearly.	5	4	2,3
	D	In how many ways can you select **2 balls** from a box containing **4 red and 3 blue** balls?	2	4	1

- End of the Question Paper -